Lab 4

3.2 $C_k = 4/\pi \int_0^{\pi/2} y(\theta)\sin(k\theta)\,d\theta$ k = odd

We know $y(\theta)$ $\begin{cases} A\sin\theta, & 0 \le \theta \le \theta_c \\ L, & \theta_c \le \theta \le \pi/2 \end{cases}$

$$C_k = \frac{4}{\pi}\left[A\int_0^{\theta_c} \sin(\theta)\sin(k\theta)\,d\theta + L\int_{\theta_c}^{\pi/2} \sin(k\theta)\,d\theta\right]$$

$k=1$, $I_1 = A\int_0^{\theta_c} \sin^2(\theta)\,d\theta = \frac{A}{2}\int_0^{\theta_c}[1-\cos(2\theta)]$

$$= \frac{A}{2}\left(\theta - \frac{\sin(2\theta)}{2}\right)\Big|_0^{\theta_c}$$ All 0 bounds go to 0

$$= \frac{A\theta_c}{2} - \frac{A\sin(2\theta_c)}{4}$$

$I_2 = L\int_{\theta_c}^{\pi/2} \sin\theta\,d\theta = L(-\cos(\theta))\Big|_{\theta_c}^{\pi/2}$ $\pi/2 \rightarrow$ goes to 0

$= L\cos(\theta_c)$

$$C_1 = \frac{4}{\pi}\left[\frac{A\theta_c}{2} - \frac{A\sin(2\theta_c)}{4} + L\cos(\theta_c)\right]$$

$-\sin(2\theta_c) = 2\sin\theta_c\cos\theta_c$ & $L = A\sin\theta_c$

$$\Rightarrow C_1 = \frac{4}{\pi}\left[\frac{A\theta_c}{2} - \frac{A}{2}\sin\theta_c\cos\theta_c + A\sin\theta_c\cos\theta_c\right]$$

$$C_1 = \frac{2A}{\pi}\left[\theta_c + \sin\theta_c\cos\theta_c\right]$$ For $k = 1$

Lab 4

Task 3.2 for $k \geq 3$ Given $\sin\theta \sin k\theta = 1/2[\cos((k-1)\theta) - \cos((k+1)\theta)]$

So $\frac{A}{2}\int_0^{\theta_c} [\cos((k-1)\theta) - \cos((k+1)\theta)]\, d\theta$

$= \frac{A}{2}\left[\frac{\sin((k-1)\theta)}{k-1} - \frac{\sin((k+1)\theta)}{k+1}\right]\Big|_0^{\theta_c}$ below 0 $= 0$ so

$I_1 = \frac{A}{2}\left[\frac{\sin((k-1)\theta_c)}{k-1} - \frac{\sin((k+1)\theta_c)}{k+1}\right]$

$I_2 \rightarrow L\int_{\theta_c}^{\pi/2} \sin(k\theta) = L\left[\frac{-\cos(k\theta)}{k}\right]\Big|_{\theta_c}^{\pi/2}$

$= \frac{L}{k}\left[-\cos\left(\frac{k\pi}{2}\right) + \cos(k\theta_c)\right]$

& Since only odd k → goes away

$I_2 = \frac{L}{k}\cos(k\theta_c)$

$C_k = \frac{4}{\pi}\left[\frac{A}{2}\left[\frac{\sin((k-1)\theta_c)}{k-1} - \frac{\sin((k+1)\theta_c)}{k+1}\right] + \frac{L}{k}\cos(k\theta_c)\right]$

$C_k = \frac{2A}{\pi}\left[\frac{\sin((k-1)\theta_c)}{k-1} - \frac{\sin((k+1)\theta_c)}{k+1}\right] + \frac{4L}{\pi k}\cos(k\theta_c)$

& $C_1 = \frac{2A}{\pi}[\theta_c + \sin\theta_c \cos\theta_c]$

So $y(t) = C_1 \sin(\omega_0 t) + C_3 \sin(3\omega_0 t) - - - -$